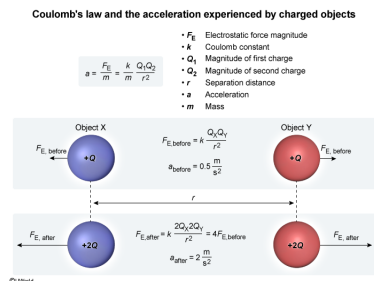


Two charged objects of equal mass, X and Y, are held a fixed distance apart. Object X is released and its instantaneous acceleration is measured as 0.50 m/s^2 . Object X is reset, and the magnitude of charge on both objects is doubled without affecting their mass. Object Y is then released. What is the instantaneous acceleration of object Y?

- ☐ A. 0.25 m/s^2 [7%]
☐ B. 0.50 m/s^2 [16%]
☐ C. 1.0 m/s^2 [31%]
☒ D. 2.0 m/s^2 [45%]

Correct
 45% answered correctly

Explanation:



Coulomb's law states that the electrostatic force between **two electric charges** is directly proportional to the product of the charge magnitude of both objects, q_1 and q_2 , and inversely proportional to the square of their separation distance r :

$$F_E = k \frac{q_1 \cdot q_2}{r^2}$$

where k is Coulomb's constant ($k = 8.9876 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$).

According to **Newton's second law of motion**, the acceleration a is directly proportional to the force F and inversely proportional to the mass m :

$$a = \frac{F}{m}$$

In this question, the mass of each object (X and Y) remains constant, but the charge is different in each experiment. When the charge on each object is doubled in the second experiment, the electrostatic force increases by a factor of 4:

$$F_{E, \text{before}} = k \frac{q_X \cdot q_Y}{r^2}$$

$$F_{E, \text{after}} = k \frac{(2 \cdot q_X)(2 \cdot q_Y)}{r^2} = 4 \cdot \left(k \frac{q_X \cdot q_Y}{r^2} \right) = 4 \cdot F_{E, \text{before}}$$

Since force and acceleration are **directly proportional** and the masses of both objects are unchanged:

$$a_{\text{after}} = \frac{F_{E, \text{after}}}{m} = 4 \cdot \frac{F_{E, \text{before}}}{m} = 4 \cdot a_{\text{before}}$$

$$a_{\text{after}} = 4 \cdot \left(0.50 \frac{\text{m}}{\text{s}^2} \right) = 2.0 \frac{\text{m}}{\text{s}^2}$$

Therefore, the acceleration will be 2.0 m/s^2 , which is four times greater than the acceleration in the first experiment.

(Choice A) Doubling the charge on each object will not reduce the acceleration by a factor of $\frac{1}{2}$ but will quadruple the acceleration due to increasing the force by a factor of 4.

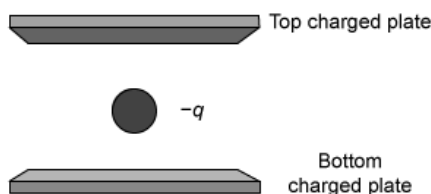
(Choice B) The charge on both objects is doubled, the force is quadrupled, and therefore the acceleration cannot remain the same.

(Choice C) If only one of the charges is doubled, then the force will be doubled and therefore the acceleration will be doubled. However, both objects' charges are doubled, not just one, causing the acceleration to be quadrupled.

Educational objective:

The electric force between two charged objects is directly proportional to the product of the magnitude of charge on both objects and inversely proportional to the square of their separation distance. The acceleration of an object is always directly proportional to the force acting on that object and inversely proportional to the mass of that object.

A droplet of oil with mass m and charge $-q$ is suspended exactly halfway between two charged plates, as shown in the figure below. The net force on the charged droplet is zero. Which of the following best describes the signs of the charges on each plate?



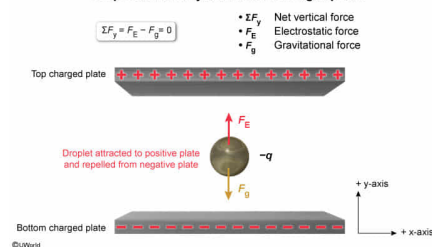
- ☐ A. Positive charge on the top plate and equal positive charge on the bottom plate [42%]
- ✓ ☒ B. Positive charge on the top plate and equal negative charge on the bottom plate [19%]
- ☐ C. Negative charge on the top plate and equal positive charge on the bottom plate [3%]
- ☐ D. Negative charge on the top plate and equal negative charge on the bottom plate [34%]

Correct

19% answered correctly

Explanation:

Zero net vertical force exerted on a droplet of oil (mass m , charge $-q$) suspended halfway between two charged plates



An **electrostatic force** F_E exists between two electrically charged objects. **Opposite charges attract and like charges repel**. In addition, an object in a gravitational field experiences a **gravitational force** F_g (ie, the **weight** of the object).

In this question, the charged droplet is positioned halfway between two plates and always experiences a **downward** F_g equal to the weight of the droplet. To remain suspended in **static equilibrium**, the plates must have a combination of charges that produce an **upward** F_E on the droplet such that the net vertical force (ie, in the y -direction) on the droplet is zero:

$$\Sigma F_y = F_E - F_g = 0$$

A positively charged top plate will attract the negatively charged droplet upward, and a negatively charged bottom plate will repel the droplet upward. The combined effect of this configuration of charges produces the required net upward electric force on the droplet. Therefore, the top plate has a positive charge and the bottom plate has a negative charge.

(Choices A and D) If the charges on both plates have **the same sign**, the net electrostatic force on the droplet will be zero due to the electric forces opposing each other. Because the force of gravity still acts downward on the droplet, the droplet cannot remain suspended.

(Choice C) This **charge configuration** produces a net downward electric force, which acts along with the downward gravitational force. Therefore, the droplet cannot remain suspended.

Educational objective:

Opposite electric charges attract and like electric charges repel. The gravitational force is attractive and acts on all objects within a gravitational field.

Time spent: 0 sec
Subject:
Physics

QID: 403939
System:
Electrostatics and Circuits